

## RESEARCH ARTICLE

Erbium(III) ion-doped borate-based glasses for 1.53  $\mu\text{m}$  broad band applicationsShaik Nayab Rasool<sup>1</sup>  | Shaik Shabeena<sup>2</sup> | Culala Rajasekharaudayar Kesavulu<sup>3</sup> | Singarapu Babu<sup>4</sup> | Asanapuram Mohan Babu<sup>5</sup>  | Vemula Venkatramu<sup>6,7</sup><sup>1</sup>Department of Physics, Rajeev Gandhi Memorial College of Engineering & Technology, Nandyal, Andhra Pradesh, India<sup>2</sup>Department of EEE, Rajeev Gandhi Memorial College of Engineering & Technology, Nandyal, Andhra Pradesh, India<sup>3</sup>Sci Laboratory, Centre for Materials for Electronics Technology (C-MET), Hyderabad, India<sup>4</sup>Ceramics and Glass Institute (CSIC), Madrid, Spain<sup>5</sup>Department of Physics, Audisankara College of Engineering & Technology, Gudur, Andhra Pradesh, India<sup>6</sup>Department of Physics, Krishna University Dr. MRAR PG Center, Nuzvid, India<sup>7</sup>Department of Physics, Yogi Vemana University, Kadapa, India

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## Abstract

Novel erbium(III) ion-doped borate-based glasses ( $\text{Er}^{3+}:\text{BCNF}$ ) by conventional melt-quenching technique were designed and synthesized. The glasses were characterized for their structural, vibrational and spectroscopic properties. The nephelauxetic ratio, bonding parameters, and Judd–Ofelt (JO) intensity parameters ( $\Omega_\lambda$ ,  $\lambda = 2, 4$  and  $6$ ) were determined by using absorption spectrum of 1 mol%  $\text{Er}_2\text{O}_3$  doped glass. These JO parameters were utilized to derive radiative properties for various excited states of erbium(III) ions. Emission cross-section for  $^4\text{I}_{13/2} \rightarrow ^4\text{I}_{15/2}$  transition of erbium(III) ions was computed through McCumber theory. The decay curves for ( $^2\text{H}_{11/2}$ ,  $^4\text{S}_{3/2}$ ) and  $^4\text{I}_{13/2}$  levels were recorded and analysed. All the results of  $\text{Er}^{3+}:\text{BCNF}$  glasses revealed that the studied glasses are efficient and thermally stable and could be suitable for display devices, optical amplification and green laser applications.

## KEYWORDS

borate-based glasses, erbium(III) ions, FTIR, Judd–Ofelt theory and optical fibre amplifiers, McCumber theory, optical studies, stimulated emission cross-section

## 1 | INTRODUCTION

In recent years, glasses doped with lanthanides (Lns) have gained much interest in the area of laser technology and in various advanced technological applications. The synthesis of erbium(III) ion-doped borate-based glasses ( $\text{Er}^{3+}:\text{BCNF}$ ) has attracted significant attention due to their superior mechanical and thermal stability, excellent thermal conductivity, cost effective synthesis methodologies when compared to synthesis of crystals.<sup>[1,2]</sup> Among the various glass based materials, Ln-doped glass materials have emerged as versatile due to their outstanding performance in numerous fields such as sensors, waveguides, television monitors, integrated optical lasers, display screens, amplifiers, optical communications and photonic devices.<sup>[3,4]</sup>

Interestingly, extensive investigations have been carried out to optimize the spectroscopic properties of Ln-doped glasses. The major focus of research regarding this class of Ln-doped glasses is to modify

several properties such as quantum efficiency, transition position and probabilities, the excited state radiative and non-radiative decay rates are critical in the design of optical devices such as optical fibres, amplifiers, lasers, colour displays, and up converters.<sup>[5–11]</sup> Surprisingly, when compared to other luminescent crystalline materials, Ln-doped glass compositions can be widely tuned, relatively simple to manufacture, cost effective, and so on.<sup>[8]</sup> There is an inhomogeneous widening of Ln emission inside the glass medium due to the distinctive distribution of ions in particular crystal field sites.<sup>[9,10]</sup> The choice of glass host is critical because the host matrix induces the crystal field, which has a significant impact on the photoluminescence (PL) of doped Ln ions that will be inserted in the glass host.<sup>[11]</sup>

Among Ln ions, erbium(III) ions are good activators due to their ladder type meta-stable levels, wide PL spans from visible to mid-infrared (IR) regions. Erbium-based solid-state lasers can operate in the eye-safe mid-IR range, demonstrating potential in medical

operations, remote sensing, atmospheric pollution monitoring, and military counter-measures.<sup>[12–15]</sup> Because of its potential green and red emission, up-conversion luminescence of erbium has been studied. The clustering of Ln ions affects the local environment and optical characteristics of the glass host at higher Ln ion concentrations. When Ln ions are collected and assembled, their interaction generally increases, resulting in luminescence quenching.<sup>[15]</sup> The greatest gain within tiny component dimensions is essential in the design of integrated light amplifiers.

Among different glass matrices, the borate glass matrix is usually preferred because of its unique features that contain wide optical transparency, low melting temperatures, high thermal strength, different coordination numbers, good chemical-strength, low-cost and enhanced Ln ion solubility.<sup>[16–19]</sup> These characteristics make the borate glass matrix a good contender for rapid ion conducting materials and other significant applications like laser hosts and biocompatibility materials. The present study is aimed (i) to prepare of new erbium(III) ion-doped sodium fluoro-borate glass ( $\text{Er}^{3+}:\text{BCNF}$ ), (ii) to identify the presence of functional groups, (iii) to determine the Judd–Ofelt (JO) parameters, (iv) to elucidate the radiative properties for BCNFER10 glass, (v) to interpret the amplification parameters for erbium(III) ion in BCNFER glasses.

## 2 | EXPERIMENTAL DETAILS

### 2.1 | Material synthesis

Erbium(III) ion-doped borate-based glasses of molar chemical composition:  $(60 - x)\text{H}_3\text{BO}_3 - 20\text{CaCO}_3 - 20\text{NaF} - x\text{Er}_2\text{O}_3$ , where  $x = 0.1, 0.3, 0.5, 1.0, 1.5$  and  $2.0$  mol%, (labelled as BCNFER01, BCNFER03, BCNFER05, BCNFER10, BCNFER15 and BCNFER20) were prepared using the traditional melt quenching method. The starting materials of boric acid ( $\text{H}_3\text{BO}_3$ ), calcium carbonate ( $\text{CaCO}_3$ ), sodium fluoride (NaF), and erbium oxide ( $\text{Er}_2\text{O}_3$ ) with 99.99% purity (Aldrich Chemical Co) were used in the glass preparation. The batch composition of high purity chemicals (about 18 g) were thoroughly grinded, the uniform mixture was collected in a porcelain crucible after being ground in an agate mortar with a pestle. The mixture was heated inside an electric furnace around  $1000\text{--}1100^\circ\text{C}$  for 50 to 60 min. and then air quenched by pouring it on uncontaminated brass mould. The glasses were annealed at a temperature ( $330^\circ\text{C}$ ) slightly less than the glass transition temperature ( $T_g$ ) to remove the thermal strains. Before evaluating the optical characteristics of the samples, they were polished.

### 2.2 | Material characterization

Refractive index ( $n = 1.648$ ) for BCNFER10 glass is measured with help of an Abbe refractometer at a wavelength of a sodium vapour lamp (589.3 nm) and 1-bromonaphthalene ( $\text{C}_{10}\text{H}_7\text{Br}$ ) as a contact fluid with a precision of 0.001. The density ( $d = 2.786$  gm/cc) was estimated by the Archimedes method using water as dipping liquid. The

X-ray diffraction (XRD) analysis of the host (BCNF) glass has been measured by using a JEOL 8530 X-ray diffractometer using  $\text{CuK}_\alpha$  radiation. Fourier-transform infrared (FTIR) spectrum was recorded in the range  $600\text{--}4000\text{ cm}^{-1}$  using Perkin-Elmer Paragon 500 FTIR spectrometer. The Perkin-Elmer Lambda-950 spectrophotometer was used to measure the optical absorption spectrum of BCNFER10 glass in the spectral range  $350\text{--}2500$  nm. The photoluminescence and lifetime data were collected on an Edinburgh FLS 980 spectrophotometer, an excitation source used as a xenon arc lamp. At room temperature, all measurements were obtained and corrected for the equipment's spectral response.

## 3 | RESULTS AND DISCUSSION

### 3.1 | Physical characteristics

Table 1 shows the physical characteristics of the glass sample (BCNFER10) under investigation. The physical parameters are calculated using relevant formulas from the literature.<sup>[20]</sup>

### 3.2 | X-ray diffraction

The XRD pattern recorded in  $10^\circ \leq \theta \leq 80^\circ$  angle range for the host glass (BCNF) is shown in Figure 1, and exhibits a broad hump at lesser angles, representing the long range structural disorder which confirm the non-crystalline nature of the glass.

**TABLE 1** Several physical properties of BCNFER10 glass

| Serial number | Physical properties                                                        | Calculated value (BCNFER10) |
|---------------|----------------------------------------------------------------------------|-----------------------------|
| 1             | Density, $d$ (gm/cm <sup>3</sup> )                                         | 2.786                       |
| 2             | Refractive index, $n$                                                      | 1.648                       |
| 3             | Erbium(III) ion concentration, $C$ (10 <sup>20</sup> ions/cc)              | 6.917                       |
| 4             | Polaron radius, $r_p$ (Å)                                                  | 4.55                        |
| 5             | Inter ionic radius, $r_i$ (Å)                                              | 11.31                       |
| 6             | Field strength, $F$ (10 <sup>16</sup> cm <sup>-2</sup> )                   | 0.144                       |
| 7             | Molar refractivity, $R_m$ (cm <sup>3</sup> )                               | 9.471                       |
| 8             | Dielectric constant, $\epsilon$                                            | 2.716                       |
| 9             | Reflection losses, $R$ (%)                                                 | 5.988                       |
| 10            | Electronic polarizability, $\alpha_e$ (10 <sup>-23</sup> cm <sup>3</sup> ) | 12.57                       |
| 11            | Average molecular weight                                                   | 72.52                       |
| 12            | Optical path length (cm)                                                   | 0.198                       |
| 13            | Molar volume, $V_m$ (cm <sup>3</sup> /mol)                                 | 26.03                       |

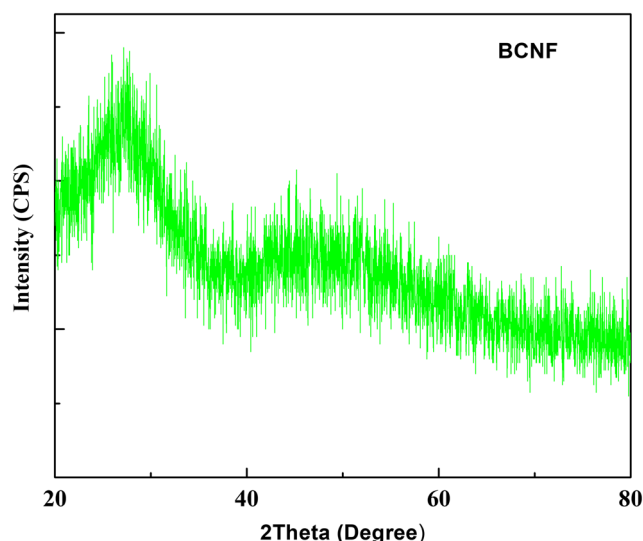


FIGURE 1 XRD profile of BCNF host glass

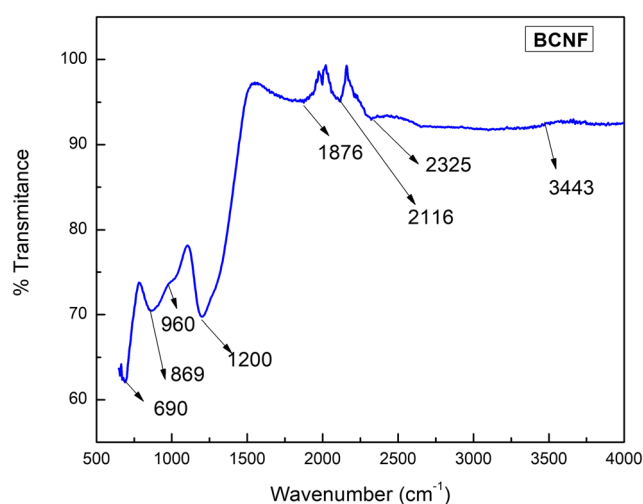


FIGURE 2 FTIR spectrum of BCNF host glass

### 3.3 | FTIR spectrum

The FTIR spectra acquired in the wavenumber range 600–4000  $\text{cm}^{-1}$  was used to estimate functional groups in the host BCNF glass (see Figure 2). As can be seen, the band at 690  $\text{cm}^{-1}$  is owing to bending vibration modes of B–O–B.<sup>[21,22]</sup> The bands at 869 and 960  $\text{cm}^{-1}$  are owing to B–O–stretching vibrations of tetrahedral  $\text{BO}_4$ .<sup>[23–25]</sup> The appearance of the peak at 1199  $\text{cm}^{-1}$  is owing to stretching B–O<sup>−</sup> in  $\text{BO}_2\text{O}^-$  units.<sup>[23–25]</sup> The peak at 1296  $\text{cm}^{-1}$  is owing to B–O stretching in  $\text{BO}_2\text{O}^-$  triangle linked with  $\text{BO}_4$  units.<sup>[23–25]</sup> The band between 1600–2900  $\text{cm}^{-1}$  belongs to hydrogen bonding in the glass matrix.<sup>[23–25]</sup> The position of the peak at 3443  $\text{cm}^{-1}$  is owing to stretching vibrations of hydroxyl clusters (O–H or H–O–H).<sup>[26]</sup>

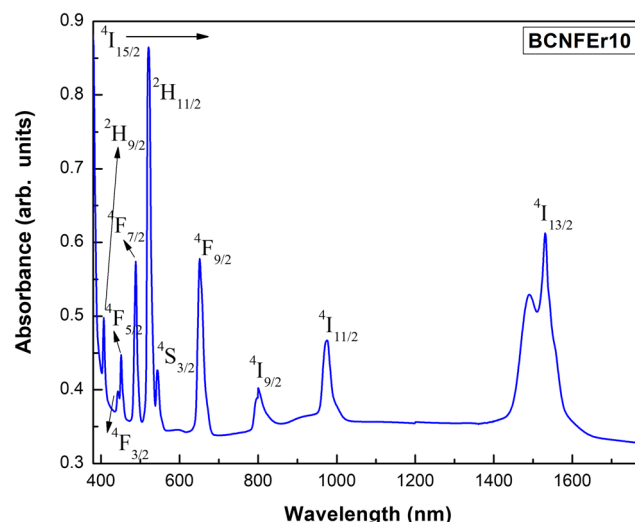


FIGURE 3 Optical absorption spectrum of BCNFEr10 glass

### 3.4 | Optical absorption spectrum, bonding and JO parameters

Figure 3 shows the ultraviolet-visible-near-infrared (UV-vis-NIR) absorption spectrum of 1.0 mol% of  $\text{Er}_2\text{O}_3$ :BCNF glass recorded at room temperature in the wavelength range 380–1780 nm. It consists of nine bands centred around 1531, 975, 800, 651, 544, 521, 488, 451 and 407 nm which are ascribed to the transitions from  $^4I_{15/2}$  lower energy level to several higher energy states of  $^4I_{13/2}$ ,  $^4I_{11/2}$ ,  $^4I_{9/2}$ ,  $^4F_{9/2}$ ,  $^4S_{3/2}$ ,  $^2H_{11/2}$ ,  $^4F_{7/2}$ ,  $^4F_{5/2}$  and  $^2H_{9/2}$ , respectively. The values of the nephelauxetic ratio ( $\beta$ ) and bonding parameter ( $\delta$ ) can be used to determine the nature of the bonding between the erbium(III) ions and their surrounding ligands. Nephelauxetic ratio ( $\beta$ ) is the ratio of the wavenumber of a certain erbium(III) ion transition in the host matrix ( $\nu_c$ ) to the same aqua-ion transition ( $\nu_a$ ).<sup>[27]</sup> Band position of the aqua-ion and transitions of the erbium(III) ions in the present glass were assigned based on the studies reported by Carnall et al.<sup>[27]</sup> The nephelauxetic ratio ( $\beta$ ) and bonding parameter ( $\delta$ ) can be calculated using the following relation<sup>[27]</sup>

$$\beta = \frac{\nu_c}{\nu_a} \quad (1)$$

$$\delta = \frac{1 - \bar{\beta}}{\bar{\beta}} \quad (2)$$

here  $\bar{\beta}$  refers to the average value of  $\beta$ .

Depending upon the sign of the  $\delta$ , the erbium(III) ion–ligand bonding may be covalent or ionic. The value of the  $\delta$  for BCNFEr10 glass was found to be 0.972 indicating that the covalent nature of the bonding between the erbium(III) ion–ligand.

The experimental and calculated oscillator strengths for various transitions of the absorption spectrum have been determined using the JO theory.<sup>[28,29]</sup> The JO theory might be used to investigate the

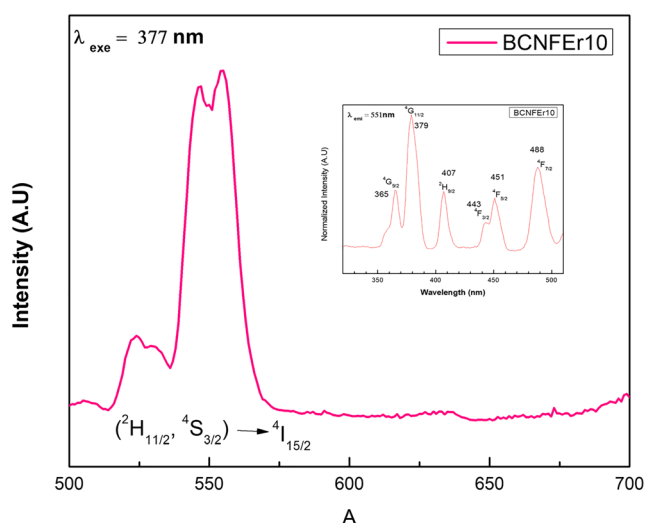
**TABLE 2** The wavelength (nm), experimental ( $f_{\text{exp}}$ ) and calculated ( $f_{\text{cal}}$ ) oscillator strengths ( $\times 10^{-6}$ ) of erbium(III) ions in BCNFER10 glass

| Level $^4I_{15/2} \rightarrow$ | $\lambda_p$ (nm) | $f_{\text{exp}}$ | $f_{\text{cal}}$ |
|--------------------------------|------------------|------------------|------------------|
| $^4I_{13/2}$                   | 1,531            | 1.42             | 1.33             |
| $^4I_{11/2}$                   | 975              | 0.64             | 0.58             |
| $^4I_{9/2}$                    | 800              | 0.52             | 0.20             |
| $^4F_{9/2}$                    | 651              | 1.60             | 1.65             |
| $^4S_{3/2}$                    | 545              | 0.51             | 0.55             |
| $^2H_{11/2}$                   | 520              | 3.82             | 2.44             |
| $^4F_{7/2}$                    | 488              | 1.44             | 1.98             |
| $^4F_{5/2} + ^4F_{3/2}$        | 451              | 1.11             | 1.04             |
| $^2G_{9/2}$                    | 407              | 0.33             | 0.78             |
| $^4G_{11/2}$                   | 377              | 2.84             | 4.32             |

$\Omega_2 = 1.10 \times 10^{-20} \text{ cm}^2$ ,  $\Omega_4 = 0.83 \times 10^{-20} \text{ cm}^2$ ,  $\Omega_6 = 1.32 \times 10^{-20} \text{ cm}^2$

**TABLE 3** Radiative parameters of the certain significant radiant levels of erbium(III) ions in BCNFER10 glass

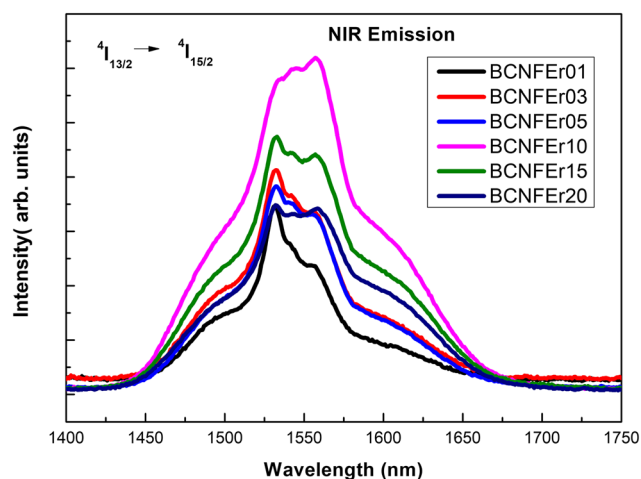
| Initial state (SLJ)       | Final state (S'L'J') | Energy gaps ( $\Delta E$ ) ( $\text{cm}^{-1}$ ) | Transition probability ( $A_R$ ) ( $\text{s}^{-1}$ ) | Branching ratio ( $\beta_R$ ) | Radiative life time ( $\tau_R$ ) ( $\mu\text{s}$ ) | Emission cross-section ( $\sigma[\lambda_p] \times 10^{-21} \text{ cm}^2$ ) |
|---------------------------|----------------------|-------------------------------------------------|------------------------------------------------------|-------------------------------|----------------------------------------------------|-----------------------------------------------------------------------------|
| $(^2H_{11/2}, ^4S_{3/2})$ | $^4F_{9/2}$          | 3,022                                           | 0.67                                                 | $\sim 0$                      | —                                                  | —                                                                           |
|                           | $^4I_{9/2}$          | 5,882                                           | 59.70                                                | 0.03                          | —                                                  | —                                                                           |
|                           | $^4I_{11/2}$         | 8,126                                           | 43.00                                                | 0.021                         | —                                                  | —                                                                           |
|                           | $^4I_{13/2}$         | 11,851                                          | 555.17                                               | 0.275                         | —                                                  | —                                                                           |
|                           | $^4I_{15/2}$         | 18,382                                          | 1361.0                                               | 0.674                         | 495                                                | 3.58                                                                        |
| $^4I_{13/2}$              | $^4I_{15/2}$         | 6,531                                           | 162                                                  | 1.00                          | 6,172                                              | 9.81                                                                        |

**FIGURE 4** Visible luminescence spectrum of BCNFER10 glass. Inset shows the excitation spectrum of BCNFER10 glass

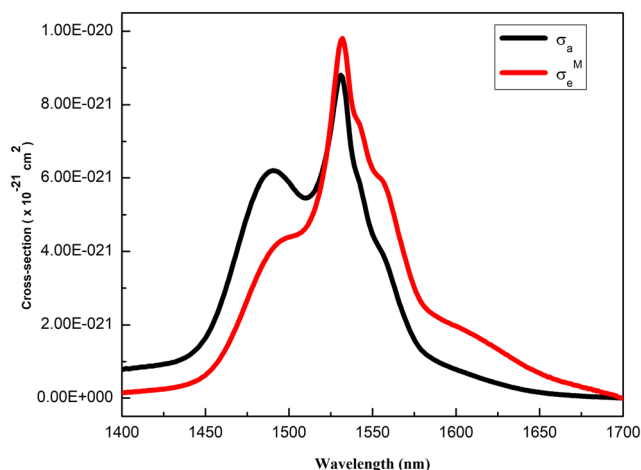
spectroscopic properties of Ln ions doped glassy materials. The measured ( $f_{\text{exp}}$ ) and calculated ( $f_{\text{cal}}$ ) oscillator strengths (see Table 2), the three JO intensity parameters  $\Omega_2 = 1.10 \times 10^{-20} \text{ cm}^2$ ,  $\Omega_4 = 0.83 \times 10^{-20} \text{ cm}^2$  and  $\Omega_6 = 1.32 \times 10^{-20} \text{ cm}^2$  are determined for the 1.0 mol%  $\text{Er}_2\text{O}_3$ -doped BCNF glass absorption spectrum, through the least square fit approach described elsewhere.<sup>[16–18]</sup> The  $\Omega_\lambda$  values follow the trend  $\Omega_4 < \Omega_2 < \Omega_6$ . The radiative transition probabilities ( $A_R$ ), radiative lifetime ( $\tau_R$ ), branching ratios ( $\beta_R$ ) and peak stimulated emission cross-section ( $\sigma[\lambda_p]$ ) for the  $(^2H_{11/2}, ^4S_{3/2}) \rightarrow ^4I_{15/2}$  and  $^4I_{13/2} \rightarrow ^4I_{15/2}$  transitions of the erbium(III) ions in borate-based glasses were computed by following the procedure described in Rasool et al.<sup>[18]</sup> and are presented in Table 3.

### 3.5 | Visible and infrared (1.54 $\mu\text{m}$ ) luminescence spectra

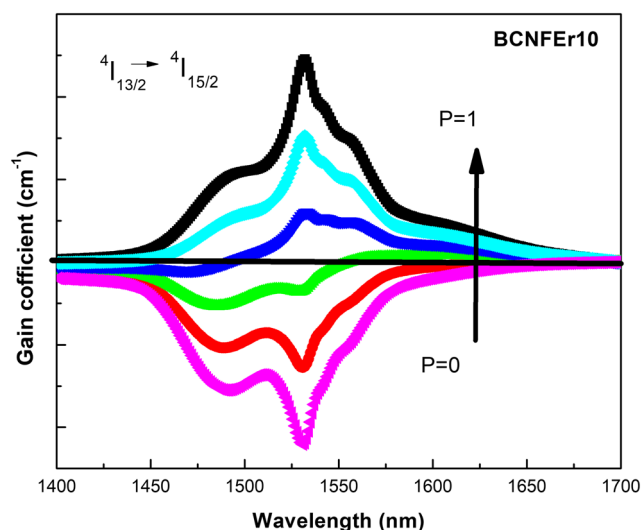
The excitation spectrum of erbium(III) ions in BCNFER10 glass is recorded by monitoring the emission at 551 nm, and the PL spectra was measured under 377 nm excitation, which are shown in Figure 4. The excitation spectrum of BCNFER10 glass exhibited six peaks at about 488, 451, 443, 407, 379 and 365 nm due to the transition of



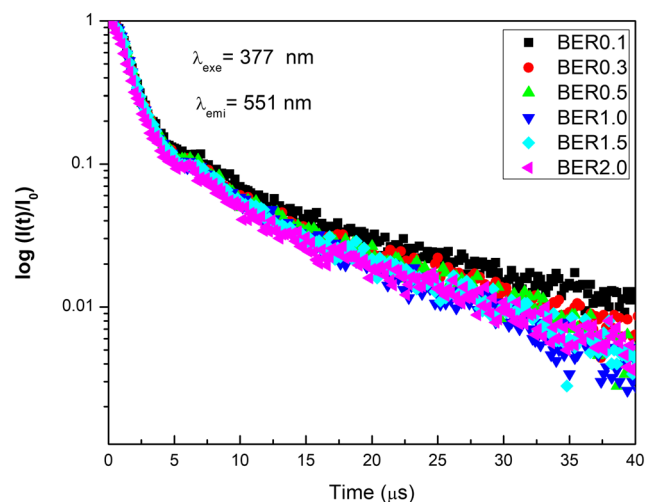
**FIGURE 5** NIR PL spectra of  $^4I_{13/2} \rightarrow ^4I_{15/2}$  transition of erbium(III) ions in BCNF



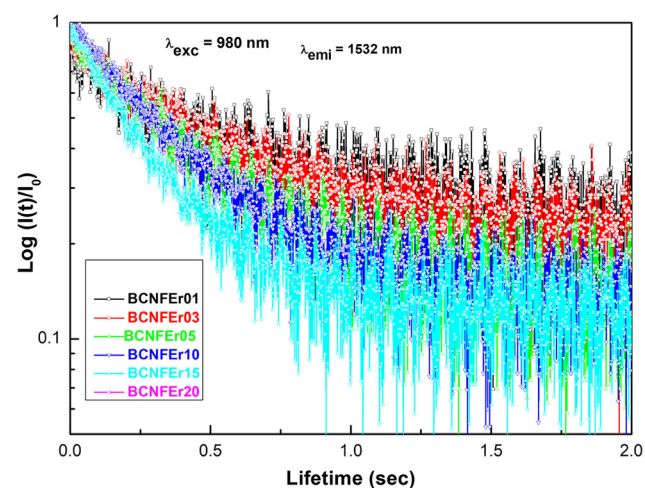
**FIGURE 6** Absorption ( $\sigma_a$ ) and emission ( $\sigma_e^M$ ) cross-sections spectra of  $^4I_{13/2} \leftrightarrow ^4I_{15/2}$  transition in BCNFEr10 glass



**FIGURE 7** Calculated gain coefficient of the BCNFEr10 glass



**FIGURE 8** Decay curves of  $^4S_{3/2} \rightarrow ^4I_{15/2}$  transition of erbium(III) ions in BCNF glasses



**FIGURE 9** Decay curves of  $^4I_{13/2} \rightarrow ^4I_{15/2}$  transition of erbium(III) ions doped BCNF glasses

erbium(III) ions from  $^4I_{15/2}$  ground level to the several higher energy levels  $^4F_{7/2}$ ,  $^4F_{5/2}$ ,  $^4F_{3/2}$ ,  $^2H_{9/2}$ ,  $^4G_{11/2}$  and  $^4G_{9/2}$  transitions, respectively. Among these the band (377 nm)  $^4I_{15/2} \rightarrow ^4G_{11/2}$  transition is more intense and aids the measurement of the visible luminescence spectrum of erbium(III) ions in BCNFEr glasses. The luminescence spectrum (Figure 4) of erbium(III) ions in BCNFEr glasses exhibited bright green emissions at 527 and 551 nm corresponding to  $^2H_{11/2} \rightarrow ^4I_{15/2}$  and  $^4S_{3/2} \rightarrow ^4I_{15/2}$  transitions, respectively.

The erbium(III) ions emission at 1.532  $\mu\text{m}$  corresponding to the  $^4I_{13/2} \rightarrow ^4I_{15/2}$  transition is the most-significant and is suitable for optical communication and laser applications. The NIR spectroscopic properties of erbium(III) ions in BCNFEr glasses have been determined as a function of erbium(III) ion concentration. The NIR emission spectra were recorded using a 980 nm diode laser to excite the glass samples, and the resulting spectra are displayed in Figure 5. The intensity of 1.532  $\mu\text{m}$  ( $^4I_{13/2} \rightarrow ^4I_{15/2}$  transition) emission was found to increase

**TABLE 4** Comparison of amplification parameters of the  $^4I_{13/2} \rightarrow ^4I_{15/2}$  transition of certain erbium(III) ions in different glass systems

| Amplification parameters of the $^4I_{13/2} \rightarrow ^4I_{15/2}$ transition                          | BCNFER10 (present work) | SLBiBER10 <sup>[16]</sup> | SANSCEr10 <sup>[33]</sup> | SBNCeEr05 <sup>[34]</sup> | BLK1.0Er <sup>[35]</sup> | PKAPbNEr10 <sup>[36]</sup> |
|---------------------------------------------------------------------------------------------------------|-------------------------|---------------------------|---------------------------|---------------------------|--------------------------|----------------------------|
| Peak position ( $\lambda_p$ , nm)                                                                       | 1532                    | 1558                      | 1535                      | 1530                      | 1533                     | 1543                       |
| Effective band widths ( $\Delta\lambda_{eff}$ , nm)                                                     | 104                     | 103                       | 53                        | 86.89                     | 81.21                    | 46                         |
| Stimulated emission cross-sections ( $\sigma[\lambda_p]$ , $\times 10^{-21}$ cm <sup>2</sup> )          | 9.81                    | 8.88                      | 9.80                      | 9.47                      | 8.22                     | 6.73                       |
| Gain bandwidths ( $(\sigma[\lambda_p] \cdot \Delta\lambda_{eff})$ , $\times 10^{-28}$ cm <sup>3</sup> ) | 1020                    | 914                       | 520                       | 820                       | 600                      | 310                        |
| Radiative lifetimes ( $\tau_{rad}$ , ms)                                                                | 6.17                    | 4.65                      | 5.18                      | 6.63                      | 7.047                    | 6.91                       |
| Experimental lifetimes ( $\tau_{exp}$ , ms)                                                             | 3.56                    | —                         | 3.83                      | 5.45                      | 5.155                    | 2.06                       |
| Quantum efficiency ( $\eta$ , %)                                                                        | 58.00                   | —                         | 74.00                     | 82.35                     | 86.00                    | 29.81                      |
| Gain per unit length ( $(\sigma[\lambda_p] \cdot \tau_{exp})$ , $\times 10^{-24}$ cm <sup>2</sup> s)    | 34.92                   | —                         | 37.57                     | 51.65                     | 42.37                    | 13.86                      |

with increase of Er<sub>2</sub>O<sub>3</sub> content up to 1.0 mol% and then there has been a decline in the intensity for 1.5 and 2.0 mol% of Er<sub>2</sub>O<sub>3</sub>-doped BCNFER glasses due to concentration quenching.

### 3.6 | McCumber theory and gain properties

McCumber theory<sup>[30]</sup> was used to calculate the stimulated emission cross-section of the  $^4I_{13/2} \rightarrow ^4I_{15/2}$  transition of erbium(III) ion. The absorption cross-section, stimulated emission cross-section for  $^4I_{13/2} \rightarrow ^4I_{15/2}$  transition of Er<sup>3+</sup> ions has been calculated.<sup>[31]</sup> Figure 6 shows the absorption and stimulated emission cross-section spectra of  $^4I_{13/2} \rightarrow ^4I_{15/2}$  transition of BCNFER10 glass as function of the wavelength. As can be seen from Figure 6 it is clear that, in the case of stimulated emission cross-section, the McCumber theory ( $9.98 \times 10^{-21}$  cm<sup>2</sup>) is in good agreement with JO theory ( $9.81 \times 10^{-21}$  cm<sup>2</sup>).

The gain properties are directly connected to the absorption and emission cross-sections. The net gain coefficient  $[G(\lambda)]$  at wavelength can be predictable using the relation available in Desurvire.<sup>[32]</sup> The gain coefficient of the  $^4I_{13/2} \rightarrow ^4I_{15/2}$  transition as a function of wavelength for BCNFER10 glass, for the various values of  $P$  (0, 0.2, 0.4, 0.6, 0.8 and 1.0), are determined and are also shown in Figure 7.

### 3.7 | Visible and infrared (1.53 $\mu$ m) decay life time measurements

The decay profiles, under 379 nm excitation, are recorded for  $^4S_{3/2}$  excited state of erbium(III) ions in BCNFER glasses and are shown in Figure 8. All of the decay curves exhibit non-exponential nature and the decay times ( $\tau_{exp}$ ) are determined, using the expression  $\tau_{exp} = \int t I(t) dt / \int I(t) dt$ . The calculated values 68, 57, 52, 42 and 40  $\mu$ s correspond to BCNFER01, BCNFER03, BCNFER05, BCNFER10, BCNFER15 and BCNFER20 glasses, respectively. The NIR decay profiles of  $^4I_{13/2} \rightarrow ^4I_{15/2}$  transition (1.53  $\mu$ m) are shown in Figure 9. The lifetime

( $\tau_{exp}$ ) values that have been found to be 5.47, 4.82, 4.33, 3.56, 2.45, and 2.44 ms for BCNFER01, BCNFER03, BCNFER05, BCNFER10, BCNFER15 and BCNFER20 glasses, respectively. The amplification parameters are listed in Table 4 along with some reported erbium(III) ion-doped systems.<sup>[16,33–36]</sup>

## 4 | CONCLUSIONS

The current article recapitulates the following

1. The transparent and moisture resistant erbium(III) ion-doped borate-based glasses were prepared.
2. The amorphous nature of the host glass was determined using XRD.
3. The FTIR spectrum revealed the main functional groups of the host glass sample.
4. The JO parameters and radiative characteristics of erbium ions in the glasses have been determined using the absorption spectrum.
5. The strong NIR luminescence at 1532 nm due to  $^4I_{13/2} \rightarrow ^4I_{15/2}$  level of erbium(III) ions are observed under the 980 nm excitation at room temperature.
6. All the obtained results indicate that the investigated glasses could be appropriate materials for long-range communications of broad-band optical amplifiers and potential candidate for lasing action.

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